



Department of Computer Science and Engineering (Data Science)

Subject: Artificial Intelligence (DJ19DSC502)

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Experiment 1

(Problem Solving)

Aim: Implement domain specific functions for given problems required for problem solving.

Theory:

There are two domain specific functions required in all problem solving methods.

1. GoalTest Function:

goalTest(State) Returns *true* if the input state is the goal state and *false* otherwise.

goalTest(State, Goal) Returns *true* if *State* matches *Goal*, and *false* otherwise.

2. MoveGen function:

Initialize set of successors C to empty set.

Add M to the complement of given state N to get new state S.

If given state has Left, then add Right to S, else add Left.

If legal(S) then add S to set of successors C.

For each other-entity E in N

make a copy S' of S,

add E to S',

If legal (S'), then add S' to C.

Return (C).

Lab Assignment to do:

Create MoveGen and GoalTest Functions for the given problems

1. Water Jug Problem

There are two jugs available of different volumes such as a 3 litres and a 7 litres and you have to measure a different volume such as 6 litre.

2. Travelling Salesman Problem

A salesman is travelling and selling his/her product to in different cities. The condition is that it has to travel each city just once.

3. 8 Puzzle Problem

An initial state is given in a 8 puzzle where one place is blank out of 9 places. You can shift this blank space and get a different state to reach to a given goal state.